

Discrete Mathematics

Week 9: Number Theory (Cont.)

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3. Primes and Greatest Common Divisors

Primes

- **Definition:** A positive integer p greater than 1 is called prime if the only positive factors of p are 1 and p .
 - A positive integer that is greater than 1 and is not prime is called composite.
 - p is composite iff there is some positive integer a such that $a \mid p$ and $1 < a < p$

3. Primes and Greatest Common Divisors

Primes

- Which of these numbers are prime?
 - 1
 - 2
 - 3
 - 4
 - 5

3. Primes and Greatest Common Divisors

Primes

- Which of these numbers are prime?
 - 6
 - 7
 - 9
 - 17
 - 51

3.1 Primes

- The study of prime numbers goes back to ancient times.
- Thousands of years ago it was known that there are infinitely many primes; the proof of this fact, found in the works of **Euclid**, is famous for its elegance and beauty.
- Primes have become essential in modern cryptographic systems (i.e. finding large primes)
 - Length of time required to factor large integers into their prime factors indicate strength of the systems.

The Fundamental Theorem of Arithmetic

Theorem 1: Every integer > 1 can be written uniquely as a prime or as the product of two or more primes, where the prime factors are written in order of non-decreasing size.

- Examples: Write the prime factorization of
 - $100 =$
 - $641 =$
 - $999 =$
 - $1024 =$

Prime Factorisation

- Every integer has a prime factorisation, but how to find it?

Problem: finding the prime factorisation of n .

- Steps:
 - Begin by dividing n by successive primes, starting with the smallest prime 2.
 - If n has a prime factor p , then p not exceeding $\sqrt{n}$ will be found. If not, then n is prime.
 - When p is found, continue by factoring n/p by successive primes, starting from p , until the factorisation has been reduced to a prime.

Prime Factorisation

- Example: Find the prime factorisation of 7007.
 - $7007 / 2 \rightarrow 7007 / 3 \rightarrow 7007 / 5$ (Not divisible)
 - $7007 / 7 = 1001$ (divisible)
 - $1001 / 7 = 143$ (divisible)
 - $143 / 7$ (Not divisible)
 - $143 / 11 = 13$ (divisible)
 - 13 is prime, the procedure is completed.
- $7007 = 7 * 1001 = 7 * 7 * 143 = \underline{7 * 7 * 11 * 13}$

3.2 Trial Division

- How to show that a given integer is prime?

Theorem 2: If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$.

- **Trial division** algorithm is a brute-force algorithm which divide n by all primes not exceeding $\sqrt{n}$ and conclude that n is prime if it is not divisible by any of these primes.

Trial Division

Example 1: Use trial division to show that 101 is prime.

3.3 The Sieve of Eratosthenes

- Find all primes not exceeding a specified positive integer by
 - Delete all the integers divisible by 2 (except 2).
 - Delete all the integers divisible by 3 (except 3).
 - ...
 - until a largest prime, a , such that $a \leq \sqrt{n}$
- Examples: Find all primes not exceeding 25.

1	2	3	4	5	6	7	8	9	10	11	12	13
14	15	16	17	18	19	20	21	22	23	24	25	

The Sieve of Eratosthenes

	2	3	4	5	6	7	8	9	10	Prime numbers
11	12	13	14	15	16	17	18	19	20	
21	22	23	24	25	26	27	28	29	30	
31	32	33	34	35	36	37	38	39	40	
41	42	43	44	45	46	47	48	49	50	
51	52	53	54	55	56	57	58	59	60	
61	62	63	64	65	66	67	68	69	70	
71	72	73	74	75	76	77	78	79	80	
81	82	83	84	85	86	87	88	89	90	
91	92	93	94	95	96	97	98	99	100	
101	102	103	104	105	106	107	108	109	110	
111	112	113	114	115	116	117	118	119	120	

3.4 Interesting Facts about Primes

1. There are an infinite number of primes - proved by **Euclid**
2. Mersenne primes are primes in the form of $2^p - 1$
 - Check out the quest to discover largest prime
<https://www.mersenne.org/>
3. The Prime Number Theorem: The ratio of the number of primes not exceeding x can be approximated by $x/\ln x$

Infinitude of Primes

Euclid
(325 B.C.E. – 265 B.C.E.)



Theorem 3: There are infinitely many primes. (Euclid)

Proof by contradiction: Assume finitely many primes: $p_1, p_2, \dots, p_n$

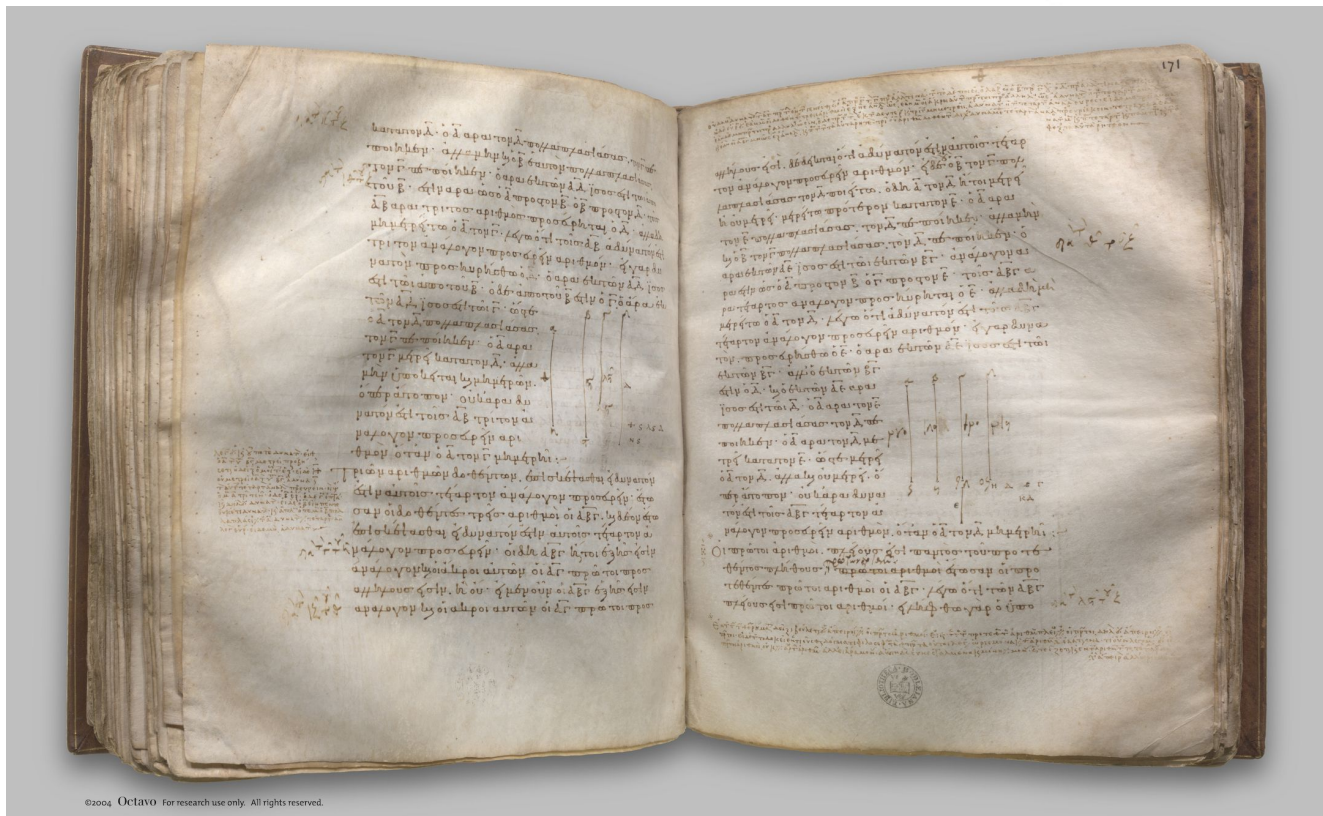
- Let $Q = p_1 p_2 \cdots p_n + 1$
- By the fundamental theorem of arithmetic, Q is prime or else it is a product of primes.
- But none of the primes p_j divides Q since if $p_j \mid Q$, then p_j divides $Q - p_1 p_2 \cdots p_n = 1$.
- Hence, there is a prime not on the list $p_1, p_2, \dots, p_n$.
- It is either Q , or if Q is composite, it is a prime factor of Q . This contradicts the assumption that $p_1, p_2, \dots, p_n$ are all the primes.
- Consequently, there are infinitely many primes.

Infinitude of Primes

Euclid
(325 B.C.E. – 265 B.C.E.)



Elements (Book IX,
Proposition 20)



The Quest to Discover Largest Prime

Table of all 52 currently-known Mersenne primes and corresponding perfect numbers

Rank	p	Mersenne prime		Perfect number		Discovery			Ref. ^[14]
		Value ↕	Digits ↕	Value ↕	Digits ↕	Date ↕	Discoverer	Method	
*	138,249,913	Lowest untested milestone ^[f]							
52 ^[g]	136,279,841	881694...871551	41,024,320	388692...008576	82,048,640	Oct 12, 2024	GIMPS / Luke Durant	LLT / PRPLL [↗] on Nvidia H100 GPU ^[h]	^[69]
51 ^[g]	82,589,933	148894...902591	24,862,048	110847...207936	49,724,095	Dec 7, 2018	GIMPS / Patrick Laroche	LLT / Prime95 on PC with Intel Core i5-4590T processor	^[67] ^[68]

Distribution of Primes

TABLE 2 Approximating $\pi(x)$ by $x / \ln x$.

x	$\pi(x)$	$x / \ln x$	$\pi(x) / (x / \ln x)$
10^3	168	144.8	1.161
10^4	1229	1085.7	1.132
10^5	9592	8685.9	1.104
10^6	78,498	72,382.4	1.084
10^7	664,579	620,420.7	1.071
10^8	5,761,455	5,428,681.0	1.061
10^9	50,847,534	48,254,942.4	1.054
10^{10}	455,052,512	434,294,481.9	1.048

3.5 Greatest Common Divisor (GCD)

- The largest integer that divides both of two integers.

Definition: Let a and b be integers, not both zero.

The largest integer d such that $d \mid a$ and also $d \mid b$ is called the *greatest common divisor* of a and b .

The greatest common divisor of a and b is denoted by $\gcd(a,b)$.

- One way to find the gcd is to find all the positive common divisors of both integers and then take the largest divisor.

Greatest Common Divisor (GCD)

Example 1: What is the greatest common divisor of 24 and 36?

Solution: The positive common divisors of 24 and 36 are

1, 2, 3, 4, 6, and 12.

Thus, $\gcd(24, 36) = \underline{\hspace{2cm}}$

Example 2: What is the greatest common divisor of 17 and 22?

Solution: $\gcd(17, 22) = \underline{\hspace{2cm}}$

Relative Prime Number

Definition: The integers a and b are relatively prime if their greatest common divisor is 1.

- Example 1: 17 and 22 are relatively prime because $\gcd(17, 22) = 1$

Definition: The integers $a_1, a_2, \dots, a_n$ are pairwise relatively prime if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

- Example 2: 10, 17, and 21 are pairwise relatively prime because $\gcd(10, 17) = 1$, $\gcd(10, 21) = 1$, and $\gcd(17, 21) = 1$.
- Example 3: 10, 19, and 24 are not pairwise relatively prime because $\gcd(10, 24) = 2$ which is > 1 .

Greatest Common Divisor (GCD) using Prime Factorisation

- Suppose the prime factorisations of a and b are:

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n} , \qquad b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n} ,$$

where each exponent is a nonnegative integer, and where all primes occurring in either prime factorisation are included in both. Then:

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)} .$$

Greatest Common Divisor (GCD) using Prime Factorisation

- This formula is valid since the integer on the right (of the equals sign) divides both a and b . No larger integer can divide both a and b .

- Example: $120 = 2^3 * 3 * 5$ and $500 = 2^2 * 5^3$

$$\begin{aligned}\gcd(120, 500) &= 2^{\min(3,2)} * 3^{\min(1,0)} * 5^{\min(1,3)} \\ &= 2^2 * 3^0 * 5^1 = 20\end{aligned}$$

- Finding the gcd of two positive integers using their **prime factorisations** is not efficient because there is no efficient algorithm for finding the prime factorisation of a positive integer.

Greatest Common Divisor (GCD) using Prime Factorisation

- Example: $\text{gcd}(120, 500)$

- $120 = 2 * 2 * 2 * 3 * 5$

- $500 = 2 * 2 * 5 * 5 * 5$

- Integers that both divides 120 and 500 are $2 * 2 * 5 = 20$

Greatest Common Divisor (GCD) Exercise

Exercise 1: What is the gcd of 84 and 96?

Exercise 2: What is the gcd of 17 and 47?

Exercise 3: What is the gcd of 2041 and 7055?

3.6 Euclidean Algorithm



- The **Euclidean algorithm** is an efficient method for computing the gcd of two integers.

Lemma 1: Let $a = bq + r$,

$\gcd(a, b) = \gcd(b, r)$ where $a > b$ and $r = a \bmod b$.

- Example: Find $\gcd(91, 287)$:
 - $287 = (91 * 3) + 14$
 - $91 = (14 * 6) + 7$
 - $14 = (7 * 2) + 0$
- $\gcd(287, 91) = \gcd(91, 14) = \gcd(14, 7) = 7$

Euclid (325 B.C.E. – 265 B.C.E.)



- **Euclid** of Alexandria was a Greek mathematician, often referred to as the "founder of geometry" or the "father of geometry"

Euclidean Algorithm



Algorithm $\text{gcd}(a, b)$: positive integers with $a > b$

$x = a, y = b$

while ($y \neq 0$):

$r = x \bmod y$

$x = y$

$y = r$

return x where $\{\text{gcd}(a, b) \text{ is } x\}$

- The number of divisions required to find the gcd is $O(\log b)$.

Euclidean Algorithm Exercise

Exercise 1: What is the gcd of 2322 and 654?

Exercise 2: What is the gcd of 27664 and 2054?

4. Applications of Congruences

- Hashing Functions (Covered in Data Structures)
- Pseudorandom Numbers
- Check Digits (Parity Check Bits, UPCs, ISBNs)

4.2 Pseudorandom Numbers

- Randomly chosen numbers are needed for many purposes, including computer simulations.
- *Pseudorandom numbers* are not truly random since they are generated by systematic methods.
- The **linear congruential method** $ax \equiv b \pmod{m}$ is one commonly used procedure for generating pseudorandom numbers.
- Four integers are needed: the **modulus** m , the **multiplier** a , the **increment** c , and **seed** x_0 , with $2 \leq a < m$, $0 \leq c < m$, $0 \leq x_0 < m$

Pseudorandom Numbers

- Python example:

```
import random
```

```
random.randint(1, 10) = 1  
random.randint(1, 10) = 3  
random.randint(1, 10) = 6  
random.randint(1, 10) = 6  
random.randint(1, 10) = 7
```

Without seed x_0

```
random.seed(9001)  
random.randint(1, 10) = 1  
random.seed(9001)  
random.randint(1, 10) = 1  
random.seed(9001)  
random.randint(1, 10) = 1
```

With seed x_0

Pseudorandom Numbers

- We generate a sequence of pseudorandom numbers $\{x_n\}$, with $0 \leq x_n < m$ for all n , by successively using the recursively defined function

$$x_{n+1} = (ax_n + c) \bmod m.$$

- If pseudorandom numbers between 0 and 1 are needed, then the generated numbers are divided by the modulus, x_n / m .

Pseudorandom Numbers Example

- Example: Find the sequence of pseudorandom numbers generated by the linear congruential method with modulus $m = 9$, multiplier $a = 7$, increment $c = 4$, and seed $x_0 = 3$.

$$x_{n+1} = (7x_n + 4) \bmod 9, \text{ with } x_0 = 3.$$

4.3 Check Digits: UPCs

- A common method of detecting errors in strings of digits is to add an extra digit at the end, which is evaluated using a function. If the final digit is not correct, then the string is assumed not to be correct.
- Example: Retail products are identified by their Universal Product Codes (UPCs). Usually these have 12 decimal digits, the last one being the check digit. The check digit is determined by the congruence:

$$3x_1 + x_2 + 3x_3 + x_4 + 3x_5 + x_6 + 3x_7 + x_8 + 3x_9 + x_{10} + 3x_{11} + x_{12} \equiv 0 \pmod{10}.$$

Check Digits: UPCs Example

- Suppose that the first 11 digits of the UPC are 79357343104. What is the check digit?

$$3 \cdot 7 + 9 + 3 \cdot 3 + 5 + 3 \cdot 7 + 3 + 3 \cdot 4 + 3 + 3 \cdot 1 + 0 + 3 \cdot 4 + x_{12} \equiv 0 \pmod{10}$$

$$21 + 9 + 9 + 5 + 21 + 3 + 12 + 3 + 3 + 0 + 12 + x_{12} \equiv 0 \pmod{10}$$

$$98 + x_{12} \equiv 0 \pmod{10}$$

$$x_{12} \equiv \underline{\hspace{1cm}} \pmod{10}$$

So, the check digit is .

Check Digits: UPCs Example

- Is 041331021641 a valid UPC?

5. Cryptography

- The subject of transforming information so that it cannot be easily recovered without special knowledge
- Number theory is the basis of many classical ciphers.
- These ciphers encrypt messages by changing each letter to a different letter. This can applies to block of letters.

Alan Turing



5.1 Classical Cryptography

Caesar Cipher

- Julius Caesar created secret messages by shifting each letter three letters forward in the alphabet (sending the last three letters to the first three letters.)
- For example, the letter B is replaced by E and the letter X is replaced by A. This process of making a message secret is an example of **encryption**.

Caesar Cipher

- Here is how the Caesar's encryption process works mathematically:
 - Replace each letter by an integer from $\mathbf{Z}_{26}$, that is an integer from 0 to 25 representing one less than its position in the alphabet.
 - For example, replace A by 0, and Z by 25.
 - The encryption function is $f(p) = (p + 3) \bmod 26$.
 - It replaces each integer p in the set $\{0,1,2,\dots,25\}$ by $f(p)$.
 - Replace each integer p by the letter with the position $p + 1$ in the alphabet.

Caesar Cipher Example

Example: What is the secret message produced from the message “WINTER IS COMING” using the Caesar cipher?

- First, replace the letters in the message with numbers.
- Replace each of these numbers p by $f(p) = (p + 3) \bmod 26$.
- Translate back to letters

Caesar Cipher Example

W I N T E R I S C O M I N G

$$f(p) = (p + 3) \bmod 26$$

Caesar Cipher Decryption

- To recover the original message, use

$$f^{-1}(p) = (p - 3) \bmod 26.$$

- So, each letter in the coded message is shifted back three letters in the alphabet, with the first three letters sent to the last three letters.
- This process of recovering the original message from the encrypted message is called **decryption**.

Shift Cipher

- The Caesar cipher is one of a family of ciphers called **shift ciphers**.
- Letters can be shifted by an integer k , with 3 being just one possibility.
- The encryption function is

$$f(p) = (p + k) \bmod 26$$

and the decryption function is

$$f^{-1}(p) = (p - k) \bmod 26$$

The integer k is called a **key**.

Shift Cipher Example

Example: Decrypt the ciphertext message “FVB ZOHSS UVA WHZZ” encrypted with the shift cipher with shift $k = 7$?

- First, translate the letters back to elements of $\mathbf{Z}_{26}$.
- Shift each of these numbers by $-k = -7 \bmod 26$.
- Translate back to letters

Shift Cipher Example

F V B Z O H S S U V A W H Z Z

ขั้นตอนแรกก็ แปลงตัวอักษร เป็นตัวเลขในเซต Z_{26} ก่อน

จากนั้นขยับหรือ shift กลับไป 7 ตัวอักษร

สุดท้ายก็ แปลงตัวเลขกลับมาเป็นตัวอักษร

More in Cybersecurity Course (for IT students)

- Affine Ciphers
- Vigenère Ciphers
- Public Key Cryptography
 - Knowing how to encrypt and two large primes allow decrypting.
- RSA Cryptosystem
- Etc.